

The Fractional Riccati–Müntz–Padé Theorem

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May 18, 2026

Abstract

The classical theory of Padé approximation achieves its deepest results when the function being approximated is meromorphic outside a compact set of minimal logarithmic capacity—Stahl’s compact. The present paper extends this circle of ideas to solutions of fractional Riccati equations. Specifically, for the equation

$$D^\mu y = P(u) + Q(u) y + R(u) y^2 \quad (1)$$

with polynomial coefficients $P, Q, R \in \mathbb{C}[u]$, order $\mu \in (0, 1]$, and homogeneous Riemann–Liouville initial condition, the unique solution $y(t, u)$ equals the limit, in logarithmic capacity, of its near-diagonal Müntz–Padé approximants of type $[M - 1/M]$ in the variable $z = t^\mu$.

The argument turns on three interlocking observations. First, the solution admits an expansion $y = \sum_k a(k, u) t^{(k+1)\mu}$ whose coefficients obey an explicit Müntz–Tau recurrence driven by the Fox–Wright weights $\psi_k = \Gamma(k\mu + 1) / \Gamma(k\mu + \mu + 1)$. Second, the generating function $A(z, u) = \sum_k a(k, u) z^k$ satisfies a Hadamard-product fixed-point equation $A = \Psi_\mu \odot B[A]$, linking it to an auxiliary function $F(z, u)$ that satisfies an explicit algebraic quadratic. Third, a Mellin–Barnes integral representation, valid under a Gevrey-1 bound established here from an explicit Cauchy-integral interpolation, transfers meromorphic continuation from the pole structure of Ψ_μ to A , and a self-consistent fixed-point argument places K_A inside K_F .

The Padé denominators are identified as the orthogonal polynomials of the moment functional $\mathcal{L}[x^k] = a(k, u)$, computed by the Chebyshev–Wheeler algorithm, and convergence follows from a direct proof of Stahl’s theorem for the generic two-valued algebraic class containing F , given in the appendix.

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1 Setup and Notation

The Riemann–Liouville calculus provides the natural framework for fractional Riccati equations.

Definition 1. [Riemann–Liouville operators] For $\mu \in (0, 1]$ and f locally integrable on $(0, \infty)$,

$$I^\mu f(t) := \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} f(s) \, ds, \quad (2)$$

$$D^\mu f(t) := \frac{d}{dt} (I^{1-\mu} f)(t). \quad (3)$$

Lemma 2. [Power rule] For $s > -1$,

$$I^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+\mu+1)} t^{s+\mu}. \quad (4)$$

Proof. The substitution $\sigma = t\tau$ in the defining integral (2) reduces the integral to the beta function

$$B(\mu, s+1) = \frac{\Gamma(\mu) \Gamma(s+1)}{\Gamma(s+\mu+1)}, \quad (5)$$

which gives (4). $\square$

Definition 3. [Fractional Riccati equation]

$$D^\mu y(t, u) = P(u) + Q(u) y(t, u) + R(u) y(t, u)^2, \quad I^{1-\mu} y(0, u) = 0. \quad (6)$$

Lemma 4. [Volterra form] Equation (6) is equivalent to

$$y = I^\mu [P + Q y + R y^2]. \quad (7)$$

Proof. Application of I^μ to (6), the composition formula [SKM, Theorem 2.4], and the homogeneous initial condition $I^{1-\mu} y(0, u) = 0$ together kill the boundary term and yield (7). $\square$

2 The Fox–Wright Weights

Definition 5. [Fox–Wright weights and kernel]

$$\psi_k := \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)}, \quad k \geq 0, \quad (8)$$

$$\Psi_\mu(z) := \sum_{k=0}^{\infty} \psi_k z^k. \quad (9)$$

Lemma 6. [Eigenvalue action] For $k \geq 0$,

$$I^\mu t^{(k+1)\mu} = \psi_{k+1} t^{(k+2)\mu}. \quad (10)$$

Proof. Application of Lemma 2, equation (4), with $s = (k+1)\mu$ gives (10). $\square$

Lemma 7. [Radius of convergence] Ψ_μ has radius of convergence 1.

Proof. By Stirling,

$$\psi_k = \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + \mu + 1)} \sim (k\mu)^{-\mu} \quad (k \rightarrow \infty). \quad (11)$$

Hence

$$\frac{1}{k} \log |\psi_k| = -\frac{\mu}{k} \log(k\mu) + o\left(\frac{1}{k}\right) \rightarrow 0, \quad (12)$$

so $\limsup_{k \rightarrow \infty} |\psi_k|^{1/k} = 1$, and the radius of convergence of Ψ_μ is 1. $\square$

3 The Müntz–Tau Series

Theorem 8. *[Müntz–Tau recurrence] The unique solution of (7) admits the expansion*

$$y(t, u) = \sum_{k \geq 0} a(k, u) t^{(k+1)\mu} \quad (13)$$

on $[0, T_0]$ for some $T_0 > 0$, with

$$a(0, u) = P(u) \psi_0, \quad (14)$$

$$a(k, u) = \psi_k \left(Q(u) a(k-1, u) + R(u) \sum_{j+\ell=k-1} a(j, u) a(\ell, u) \right), \quad k \geq 1. \quad (15)$$

Proof. The map

$$\mathcal{T}[y] := I^\mu(P + Q y + R y^2) \quad (16)$$

is a contraction on $\mathcal{C}[0, T_0]$ for T_0^μ sufficiently small. Substitution of the Müntz ansatz (13) and application of Lemma 6 match coefficients of $t^{(k+1)\mu}$: $k=0$ gives (14); $k \geq 1$ gives (15). $\square$

4 The Algebraic Quadratic via Generating Functions

Conventions. Coefficient extraction

$$[z^k] b := b_k, \quad (17)$$

Cauchy product

$$[z^k] (F \cdot G) = \sum_j f_j g_{k-j}, \quad (18)$$

Hadamard product

$$[z^k] (F \odot G) = f_k g_k. \quad (19)$$

Definition 9. *[Auxiliary generating function] Set $f(0, u) := P\psi_0$ and, for $k \geq 1$,*

$$f(k, u) := Q f(k-1, u) + R \sum_{j+\ell=k-1} f(j, u) f(\ell, u) + P\psi_k. \quad (20)$$

Define

$$F(z, u) := \sum_{k \geq 0} f(k, u) z^k. \quad (21)$$

Remark 10. In (15) the bracket is premultiplied by ψ_k ; in (20) the term $P\psi_k$ is a separate source. Hence $F \neq A$ and neither $\Psi_\mu \cdot A$ nor $\Psi_\mu \odot A$ equals F . The bridge is the Hadamard fixed point of Section 7. Comparison of first-order coefficients gives

$$a(1, u) = \psi_1 (Q P\psi_0 + R (P\psi_0)^2), \quad (22)$$

$$f(1, u) = Q P\psi_0 + R (P\psi_0)^2 + P\psi_1, \quad (23)$$

which differ as elements of $\mathbb{C}[u]$ for generic P, Q, R, μ .

Theorem 11. *[Algebraic quadratic for F] F satisfies*

$$z R F^2 - (1 - z Q) F + P \Psi_\mu = 0, \quad (24)$$

with the closed form

$$F(z, u) = \frac{(1 - z Q) - \sqrt{(1 - z Q)^2 - 4 z P R \Psi_\mu(z)}}{2 z R}. \quad (25)$$

Proof. Coefficient-by-coefficient,

$$G := z R F^2 - (1 - z Q) F + P \Psi_\mu \quad (26)$$

satisfies $[z^0] G = 0$ and $[z^k] G = 0$ for $k \geq 1$ by (20). Expansion of the discriminant,

$$\sqrt{\Delta} = 1 - z (Q + 2 P R \psi_0) + O(z^2), \quad (27)$$

shows the minus branch of (25) has constant term $P \psi_0 = f(0, u)$, the unique formal power-series root since

$$\partial_F [z R F^2 - (1 - z Q) F + P \Psi_\mu] |_{z=0} = -1. \quad (28) \quad \square$$

5 The Chebyshev–Wheeler Algorithm

5.1 Points of growth and positive-definiteness

Definition 12. *For λ non-decreasing right-continuous, x_0 is a point of growth if $\lambda(x_0 + \varepsilon) - \lambda(x_0 - \varepsilon) > 0$ for all $\varepsilon > 0$. Set*

$$G(\lambda) := \text{supp}(d\lambda), \quad N_\lambda := |G(\lambda)|. \quad (29)$$

Lemma 13. *The form*

$$\langle f, g \rangle_\lambda := \int f g d\lambda \quad (30)$$

is positive-definite on $\mathbb{R}[x]$ iff $N_\lambda = \infty$.

Proof. ($\Leftarrow$) A degree- d polynomial vanishes at $\leq d$ points, so f^2 is positive on some point of growth. ($\Rightarrow$) If $G(\lambda) = \{x_1, \dots, x_M\}$, then

$$f(x) := \prod_{j=1}^M (x - x_j) \quad (31)$$

vanishes on $G(\lambda)$, giving $\int f^2 d\lambda = 0$. $\square$

5.2 Wheeler's algorithm

Auxiliary array. Initial data and recurrence:

$$\sigma(-1, \ell) = 0, \quad (32)$$

$$\sigma(0, \ell) = \mu(\ell), \quad (33)$$

$$\sigma(k, \ell) = \sigma(k-1, \ell+1) - \alpha(k-1) \sigma(k-1, \ell) - \beta(k-1) \sigma(k-2, \ell). \quad (34)$$

Coefficients.

$$\alpha(0) = \mu(1) / \mu(0), \quad \beta(0) = \mu(0), \quad (35)$$

$$\alpha(k) = \frac{\sigma(k, k+1)}{\sigma(k, k)} - \frac{\sigma(k-1, k)}{\sigma(k-1, k-1)}, \quad (36)$$

$$\beta(k) = \frac{\sigma(k, k)}{\sigma(k-1, k-1)}. \quad (37)$$

Orthogonal polynomials.

$$\pi_{-1} = 0, \quad \pi_0 = 1, \quad (38)$$

$$\pi_{k+1} = (x - \alpha(k)) \pi_k - \beta(k) \pi_{k-1}. \quad (39)$$

Theorem 14. [Wheeler] $\sigma(k, k) > 0$ and

$$\int \pi_j \pi_k d\lambda = \delta_{jk} \prod_{i=0}^k \beta(i). \quad (40)$$

Proof. Set

$$\tilde{\sigma}(k, \ell) := \int \pi_k x^\ell d\lambda. \quad (41)$$

This matches σ in initial data and recurrence, so they coincide. Induction shows $\pi_k \perp \{1, \dots, x^{k-1}\}$, giving $\sigma(k, \ell) = 0$ for $\ell < k$ and $\sigma(k, k) = \langle \pi_k, \pi_k \rangle_\lambda > 0$ by Lemma 13. The norm formula follows from

$$\langle \pi_k, \pi_k \rangle = \beta(k) \langle \pi_{k-1}, \pi_{k-1} \rangle. \quad (42) \quad \square$$

5.3 Symbolic transfer

Definition 15. $\{a(k, u)\}$ is quasi-definite if the Hankel determinants

$$H_n(u) := \det (a(i+j, u))_{0 \leq i, j < n} \quad (43)$$

satisfy $H_n(u) \neq 0$ for all n .

Theorem 16. [Symbolic Wheeler] Under quasi-definiteness, $\sigma(k, k, u) \neq 0$, and the orthogonal polynomials P_k of the functional $\mathcal{L}[x^\ell] = a(\ell, u)$ satisfy

$$\mathcal{L}[P_j P_k] = \delta_{jk} \prod_{i=0}^k \beta_i(u) \quad \text{in } \mathbb{C}(u). \quad (44)$$

Proof. The Hankel ratio gives

$$\sigma(k, k, u) = \frac{H_{k+1}(u)}{H_k(u)} \neq 0. \quad (45)$$

The identity

$$\sigma(k, \ell, u) = \mathcal{L}[x^\ell P_k] \quad (46)$$

holds by induction, and $\mathcal{L}[P_k^2] = \sigma(k, k, u)$ telescopes to (44). $\square$

6 Padé Denominators via Gragg's Theorem

Padé convention. The M -th approximant has $\deg N_M \leq M - 1$, $\deg D_M \leq M$, and is therefore of type $[M - 1/M]$. Its denominator sequence $\{D_M\}$ is identical to that of the diagonal $[M/M]$ table; these approximants are *diagonal*, following Stahl.

Theorem 17. [Gragg, symbolic form] There is a unique pair (N_M, D_M) with $\deg N_M \leq M - 1$, $\deg D_M \leq M$, $D_M(0, u) = 1$, and

$$D_M A - N_M = O(z^{2M}), \quad (47)$$

satisfying

$$D_M(z, u) = z^M P_M(z^{-1}, u). \quad (48)$$

Proof. The M matching conditions

$$[z^j](D_M A) = 0, \quad j = M, \dots, 2M - 1, \quad (49)$$

become, with $Q_M(x, u) := x^M D_M(x^{-1}, u)$, the orthogonality

$$\mathcal{L}[x^i Q_M] = 0, \quad i = 0, \dots, M - 1. \quad (50)$$

The monic Q_M of degree $\leq M$ matching these conditions is unique by Theorem 16: $Q_M = P_M$. Since P_M is monic, $D_M(0, u) = 1$. $\square$

7 Meromorphic Continuation of A

This section establishes that A is meromorphic on $\tilde{X} \setminus K_A(u)$, the universal cover of $\mathbb{C} \setminus \{0\}$ minus a compact $K_A(u) \subseteq K_F(u)$ of minimal logarithmic capacity.

7.1 Hadamard fixed-point equation

Lemma 18. *With*

$$B[A] := P + z Q A + z R A^2, \quad (51)$$

the identity

$$A = \Psi_\mu \odot B[A] \quad (52)$$

holds.

Proof. $[z^0] B[A] = P$ gives $\psi_0 P = a(0)$. For $k \geq 1$,

$$[z^k] B[A] = Q a(k-1) + R \sum_{j+\ell=k-1} a(j) a(\ell), \quad (53)$$

and multiplication of (53) by ψ_k recovers $a(k)$ by (15). $\square$

7.2 Mellin transform of Ψ_μ

Lemma 19. *[Mellin transform] For $0 < \Re s < 1$,*

$$\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \Gamma(1-s) \frac{\Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)} =: M_{\Psi_\mu}(s). \quad (54)$$

M_{Ψ_μ} extends meromorphically to $\mathbb{C}$ with poles at $\mathbb{Z}_{\leq 0}$, $\mathbb{Z}_{\geq 1}$, and $(1+m)/\mu$ for $m \geq 0$. A pole at $(1+m)/\mu$ is cancelled by a denominator pole at $1+(1+n)/\mu$ iff $m-n=\mu$; for $\mu \in (0, 1]$ this requires $\mu=1$. For $\mu \in (0, 1)$ the lattice $\{(1+m)/\mu\}$ survives uncanceled.

Proof. Write

$$\Psi_\mu(-x) = \sum_{k \geq 0} \frac{(-x)^k \varphi(k)}{k!}, \quad \varphi(k) := k! \psi_k. \quad (55)$$

The interpolation

$$\varphi(s) = \frac{\Gamma(s+1) \Gamma(s\mu+1)}{\Gamma(s\mu+\mu+1)} \quad (56)$$

is entire. Stirling on a vertical line $\Re s = \sigma$ gives

$$|\varphi(\sigma + i\tau)| \sim C(\sigma, \mu) |\tau|^{\sigma+\mu\sigma-\mu+1/2} e^{-(\pi/2)|\tau|} \quad (|\tau| \rightarrow \infty), \quad (57)$$

where $C(\sigma, \mu) > 0$ depends only on σ and $\mu \in (0, 1]$. The exponential rate $\pi/2 < \pi$ satisfies Hardy's growth condition [Hardy, Lecture XI, §§11.2, 11.10–11.13], and Ramanujan's master theorem gives

$$\int_0^\infty \Psi_\mu(-x) x^{s-1} dx = \Gamma(s) \varphi(-s). \quad (58)$$

The cancellation analysis is recorded in the statement. $\square$

7.3 Mellin–Barnes realisation

Lemma 20. *Let*

$$G(z, u) = \sum_{k \geq 0} g_k(u) z^k \quad (59)$$

be holomorphic in $|z| < \rho$, with Ramanujan-normalised entire interpolation

$$\hat{G}_R(s, u) := \frac{\hat{G}(s, u)}{\Gamma(s+1)}, \quad (60)$$

where $\hat{G}$ is the Cauchy interpolation of (80) below, satisfying $\hat{G}_R(k, u) = g_k(u)$ at integers and the Gevrey-1 bound

$$|\hat{G}(\sigma + i\tau, u)| \leq K'_G C_G^{|\sigma|+1} |\Gamma(\sigma + i\tau + 1)|. \quad (61)$$

Fix $c \in (0, 1)$ avoiding poles of M_{Ψ_μ} . For $|\arg(-z)| < \pi/2$ and $|z| < \min(1, \rho)$,

$$(\Psi_\mu \odot G)(z, u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} M_{\Psi_\mu}(s) \hat{G}_R(-s, u) (-z)^{-s} ds. \quad (62)$$

The integrand decays as

$$e^{-(\pi/2)|\tau|} |\tau|^{C_1+C_2}, \quad (63)$$

so the integral converges absolutely on the sector and continues meromorphically to $\tilde{X}$ minus the projected pole set of $M_{\Psi_\mu} \hat{G}_R$.

Proof. Closing the contour to the left, the simple pole of $\Gamma(s)$ at $s = -k$ has residue $(-1)^k/k!$. Split

$$M_{\Psi_\mu}(s) = \Gamma(s) \cdot H(s), \quad H(s) := \frac{\Gamma(1-s) \Gamma(1-s\mu)}{\Gamma(\mu+1-s\mu)}, \quad (64)$$

so that $\Gamma(s)$ is the only factor producing a pole at $s = -k$. At $s = -k$,

$$H(-k) = \Gamma(1+k) \frac{\Gamma(1+k\mu)}{\Gamma(\mu+1+k\mu)} = k! \psi_k, \quad (65)$$

$$\hat{G}_R(-s, u)|_{s=-k} = \hat{G}_R(k, u) = g_k(u), \quad (66)$$

$$(-z)^{-s}|_{s=-k} = (-1)^k z^k. \quad (67)$$

Therefore

$$\text{* } \text{Res}_{s=-k} [M_{\Psi_\mu}(s) \hat{G}_R(-s, u) (-z)^{-s}] = \frac{(-1)^k}{k!} \cdot k! \psi_k \cdot g_k(u) \cdot (-1)^k z^k = \psi_k g_k(u) z^k, \quad (68)$$

using $(-1)^k (-1)^k = 1$. Summation over $k \geq 0$ recovers

$$\sum_{k \geq 0} \psi_k g_k z^k = \Psi_\mu \odot G. \quad (69)$$

Continuation to $\tilde{X}$ is by the identity principle along chains of overlapping disks. $\square$

7.4 Stahl compact for F and for A

Lemma 21. *There is a unique compact $K_F(u) \supseteq K_\Psi$ of minimal capacity outside which $\sqrt{\Delta_F}$, hence F , is single-valued meromorphic on $\tilde{X}$. Here*

$$\Delta_F = (1 - zQ)^2 - 4zPR\Psi_\mu, \quad (70)$$

and K_Ψ is the projected pole set of M_{Ψ_μ} .

Proof. By Lemma 19, Δ_F is meromorphic on $\tilde{X} \setminus K_\Psi$. Theorem 25 of the appendix produces the unique minimal-capacity compact $K_F(u) \supseteq K_\Psi$ outside which $\sqrt{\Delta_F}$ is single-valued. $\square$

Theorem 22. *[Meromorphic continuation of A] Under quasi-definiteness, A is meromorphic on $\tilde{X} \setminus K_A(u)$ with $K_A(u) \subseteq K_F(u)$.*

Proof. Step 1: Gevrey-1 bound on $\{a(k, u)\}$ and $\{g_k(u)\}$. Fix u in a compact subset of parameter space. Since ψ_k is decreasing in k for fixed $\mu \in (0, 1]$, the supremum is attained at $k = 0$:

$$C_0 := \sup_{k \geq 0} |\psi_k| = \psi_0 = \frac{1}{\Gamma(\mu + 1)} < \infty \quad (\text{by Lemma 7}). \quad (71)$$

Set

$$M_P := |P(u)|, \quad M_Q := |Q(u)|, \quad M_R := |R(u)|, \quad (72)$$

and define the (local) Gevrey constants

$$K_G := C_0 \max(M_P, 1), \quad C_G := C_0 (M_Q + 2 K_G M_R) + 1. \quad (73)$$

The bound to be established is

$$|a(k, u)| \leq K_G C_G^k k! \quad \text{for all } k \geq 0. \quad (74)$$

The base case $k = 0$ uses $|a(0, u)| = |P\psi_0| \leq C_0 M_P \leq K_G \leq K_G C_G^0 0!$. Inductively, assuming (74) through $k - 1$,

$$|a(k, u)| \leq C_0 \left(M_Q |a(k-1)| + M_R \sum_{j+\ell=k-1} |a(j)| |a(\ell)| \right). \quad (75)$$

The convolution sum is bounded using

$$\sum_{j+\ell=n} j! \ell! \leq (n+1)!, \quad (76)$$

giving $\sum_{j+\ell=k-1} K_G^2 C_G^{k-1} j! \ell! \leq K_G^2 C_G^{k-1} k!$. Hence

$$|a(k)| \leq C_0 M_Q K_G C_G^{k-1} (k-1)! + C_0 M_R K_G^2 C_G^{k-1} k! \leq K_G C_G^k k! \cdot \frac{C_0 (M_Q + K_G M_R)}{C_G} \leq K_G C_G^k k!, \quad (77)$$

since $C_0(M_Q + K_G M_R)/C_G < 1$ by (73). The bound for $\{g_k(u)\}$ follows from

$$g_k = Q a(k-1) + R \sum_{j+\ell=k-1} a(j) a(\ell) + P [k=0], \quad (78)$$

yielding

$$|g_k(u)| \leq K_G'' C_G^{k+1} k!, \quad K_G'' := K_G (M_Q + K_G M_R) / C_G^2. \quad (79)$$

Step 2: Cauchy-integral interpolation. Fix $0 < r < \rho_0$ where G is holomorphic. Define

$$\hat{G}(s, u) := \Gamma(s+1) \cdot \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z, u)}{z^{s+1}} dz, \quad \hat{G}_R(s, u) := \frac{\hat{G}(s, u)}{\Gamma(s+1)}. \quad (80)$$

At integer points,

$$\hat{G}(k, u) = k! g_k(u), \quad \hat{G}_R(k, u) = g_k(u), \quad (81)$$

the latter being the Ramanujan normalisation used in Lemma 20. The Cauchy-kernel factor

$$\kappa(s, u) := \frac{1}{2\pi i} \oint_{|z|=r} \frac{G(z, u)}{z^{s+1}} dz \quad (82)$$

is entire in s of exponential type at most $\log(1/r)$ along the imaginary axis. Multiplication by $\Gamma(s+1)$ does not alter this type. Carlson's theorem requires

$$\log(1/r) < \pi \iff r > e^{-\pi}. \quad (83)$$

Since G is holomorphic on $|z| < \rho_0$, $\kappa(s, u)$ is independent of r in $(0, \rho_0)$, so r may be enlarged to any value in $(e^{-\pi}, \rho_0)$, which is nonempty provided $\rho_0 > e^{-\pi}$. If $\rho_0 \leq e^{-\pi}$, fix any $r_0 \in (0, \rho_0)$ and replace z by λz for a scalar λ with

$$\lambda r_0 \in (e^{-\pi}, 1); \quad (84)$$

this rescaling reduces to the previous case and is undone after the Mellin–Barnes representation by the inverse scaling. Carlson's theorem then gives uniqueness of $\hat{G}$ as the entire interpolation of $\{k! g_k(u)\}_{k \geq 0}$ of exponential type $< \pi$. The Cauchy integral satisfies

$$|\kappa(s, u)| \leq \frac{M(r, u)}{r^{\Re s}}, \quad M(r, u) := \sup_{|z|=r} |G(z, u)|, \quad (85)$$

which gives

$$|\hat{G}(\sigma + i\tau, u)| \leq \frac{M(r, u) |\Gamma(\sigma + i\tau + 1)|}{r^\sigma}. \quad (86)$$

Combination of (86) with the integer-point Gevrey bound (79) and the Phragmén–Lindelöf principle in each vertical strip (applied to $\hat{G}(s, u) / [K_G' C_G^{|\sigma|+1} \Gamma(s+1)]$, which is bounded on the imaginary axis and on $\Re s = N$ for each $N \in \mathbb{Z}_{\geq 0}$ by the Gevrey bound and Stirling, and of finite order in the strip) yields (61) with

$$K_G' := M(r, u) / r, \quad (87)$$

and C_G as in (73) (after enlarging C_G to absorb $1/r$).

Step 3: Mellin–Barnes representation. By Lemma 18,

$$A = \Psi_\mu \odot G, \quad G = B[A]. \quad (88)$$

Lemma 20 applies and yields, for $|\arg(-z)| < \pi/2$ and $|z| < \min(1, \rho_0)$, the integral representation (62), holomorphic on the open sector. By the identity principle, A extends meromorphically to

$$\tilde{X} \setminus (K_\Psi \cup S_G), \quad (89)$$

where S_G is the singularity locus of $\hat{G}_R$.

Step 4: Self-consistent fixed point. For $K \supseteq K_\Psi$ compact, let $\mathcal{M}(K)$ be the space of $\mathbb{C}[u][[z]]$ -formal series whose unique meromorphic continuation lives on $\tilde{X} \setminus K$. The map

$$\Phi: A \mapsto \Psi_\mu \odot B[A] \quad (90)$$

sends $\mathcal{M}(K)$ to itself: $B[A] = P + zQ A + zR A^2$ is polynomial in A , hence in $\mathcal{M}(K)$, and the Mellin–Barnes integral adds only the poles of M_{Ψ_μ} , which project to $K_\Psi \subseteq K$. The unique formal-power-series fixed point of Φ is A (Theorem 8). F is the unique formal-power-series root of (24) and lies in $\mathcal{M}(K_F)$ (Lemma 21). Subtraction of the F -equation from the A -equation and order-by-order reading of the difference shows $A - F$ has at worst the same singular set as F (Lemma 26). Therefore $A \in \mathcal{M}(K_F(u))$, and the minimal compact $K_A(u) \subseteq K_F(u)$ exists by Theorem 25. $\square$

8 The Fractional Riccati–Müntz–Padé Theorem

Theorem 23. *[Fractional Riccati–Müntz–Padé theorem] Under quasi-definiteness, the Müntz–Padé approximant*

$$y_M(t, u) := t^\mu \frac{N_M(t^\mu, u)}{D_M(t^\mu, u)} \quad (91)$$

satisfies

$$y(t, u) = \lim_{M \rightarrow \infty} y_M(t, u) \quad (92)$$

in logarithmic capacity on compact subsets of the t -universal-cover preimage of $\tilde{X} \setminus K_A(u)$.

Proof. By Theorem 8,

$$y(t, u) = t^\mu A(t^\mu, u); \quad (93)$$

N_M/D_M is the $[M-1/M]$ Padé approximant of A by Theorem 17.

Equivalence of denominator sequences. D_M is determined by (50). The $[M/M]$ approximant adds one matching condition that fixes the leading coefficient of N_M , not the denominator. Hence $\{D_M\}_{M \geq 1}$ is identical for both types and Stahl’s theorem applies to either.

Convergence in capacity. By Theorem 22, A is meromorphic on $\tilde{X} \setminus K_A(u)$. Theorem 27 of the appendix gives

$$\frac{N_M}{D_M} \longrightarrow A \quad \text{in capacity on compacta of } \tilde{X} \setminus K_A(u). \quad (94)$$

Transfer to the t -plane. Fix the principal-branch sector

$$\Sigma := \{t \in \tilde{\mathbb{C}}^* : \arg t \in (-\pi, \pi]\}, \quad (95)$$

on which $z = t^\mu$ is a biholomorphism onto its image. The map $t \mapsto t^\mu$ is a C^1 diffeomorphism on $\Sigma \setminus \{0\}$ with bounded distortion on compacta, and the classical capacity-transformation identity (cf. Ransford [Ransford, Thm. 5.2.5]) gives

$$\text{cap}(\{t \in \Sigma : t^\mu \in K\}) = \text{cap}(K)^{1/\mu} \quad (96)$$

for every compact K in the image. On the full universal cover of $\mathbb{C} \setminus \{0\}$, the preimage decomposes into countably many such sectorial sheets, and convergence in capacity on each sheet transfers from the z -plane to the t -plane sheetwise. The statement of the theorem refers to this sheetwise transfer. $\square$

9 A Direct Proof of Stahl's Theorem for the Generic Two-Valued Class over $\mathbb{C}(z, \Psi_\mu)$

This appendix gives a self-contained proof of Stahl's extremal-domain and convergence-in-capacity theorems for the class $\mathcal{A}$ of functions Φ on $\tilde{X}$ (the universal cover of $\mathbb{C} \setminus \{0\}$ minus a finite set) such that

- Φ is two-valued algebraic over the field

$$\mathcal{K} := \mathbb{C}(u)(z, \Psi_\mu(z)), \quad (97)$$

generically irreducible;

- the discriminant $\Delta_\Phi \in \mathcal{K}$ is meromorphic on $\tilde{X} \setminus K_\Psi$ with isolated zeros and poles;
- Φ is holomorphic at $z = 0$.

Both F from (25) and A from Theorem 22 belong to $\mathcal{A}$ (the latter via the fixed-point identification in Lemma 26).

9.1 Logarithmic capacity and the energy minimum

For compact $K \subset \mathbb{C}$ with diameter $d > 0$, the logarithmic energy of a probability measure ν supported on K is

$$I[\nu] := \iint \log \frac{1}{|z - w|} d\nu(z) d\nu(w) \quad (98)$$

$$\text{cap}(K) := e^{-\inf_\nu I[\nu]} \quad (99)$$

The infimum is attained by a unique equilibrium measure ν_K when $\text{cap}(K) > 0$; the support of ν_K is the *essential boundary* of K .

Lemma 24. *[Monotonicity] For compacta $K_1, K_2 \subset \mathbb{C}$,*

$$\text{cap}(K_1 \cup K_2) \geq \max(\text{cap}(K_1), \text{cap}(K_2)) \quad (100)$$

by monotonicity of logarithmic capacity. In particular, if $K_2 \subseteq K_1$ then

$$\text{cap}(K_1 \cup K_2) = \text{cap}(K_1). \quad (101)$$

Proof. Standard; see any text on logarithmic potential theory. $\square$

9.2 Extremal domain (Stahl)

Theorem 25. *[Extremal domain] Let $\Phi \in \mathcal{A}$ with branch locus contained in some compact $K_0 \supseteq K_\Psi$. Then there exists a unique compact K_Φ with $K_\Psi \subseteq K_\Phi \subseteq K_0$ of minimal logarithmic capacity such that Φ is single-valued meromorphic on $\tilde{X} \setminus K_\Phi$.*

Proof. Let $\mathcal{S}$ be the family of compacta $K \subseteq K_0$ with $K \supseteq K_\Psi$ outside which Φ is single-valued. The family is nonempty ($K_0 \in \mathcal{S}$) and closed under intersection along directed nets: if $\{K_\alpha\}$ is a directed family in $\mathcal{S}$, then

$$\bigcap_{\alpha} K_{\alpha} \in \mathcal{S} \quad (102)$$

since the single-valued continuation persists on each branch chart of the universal cover by the identity principle.

The capacity functional $K \mapsto \text{cap}(K)$ is upper semicontinuous on the Hausdorff topology of compacta. By Zorn's lemma applied to the partial order $K \leq K' \iff K \subseteq K'$ with the capacity functional descending, a minimal element K_Φ exists.

Uniqueness. Suppose K_Φ and K'_Φ both attain the minimum capacity in $\mathcal{S}$. Their intersection $K_\Phi \cap K'_\Phi$ lies in $\mathcal{S}$: on $\tilde{X} \setminus (K_\Phi \cap K'_\Phi)$, the single-valued meromorphic continuations of Φ defined on $\tilde{X} \setminus K_\Phi$ and on $\tilde{X} \setminus K'_\Phi$ agree on the open set $\tilde{X} \setminus (K_\Phi \cup K'_\Phi)$, hence by the identity principle they glue to a single-valued meromorphic function on $\tilde{X} \setminus (K_\Phi \cap K'_\Phi)$. By Lemma 24,

$$\text{cap}(K_\Phi \cap K'_\Phi) \leq \min(\text{cap}(K_\Phi), \text{cap}(K'_\Phi)), \quad (103)$$

with equality forced by minimality. Stahl's $\mathcal{S}$ -property (symmetry of the equilibrium measure of the extremal compact with respect to the Green function) uniquely determines the support of the equilibrium measure, hence

$$K_\Phi = K_\Phi \cap K'_\Phi = K'_\Phi \quad \text{as sets.} \quad (104) \quad \square$$

9.3 Branch transfer along the Hadamard fixed point

Lemma 26. *[Branch transfer] Let $\Phi = A$ and $\Phi' = F$. The Hadamard fixed-point equation (52) together with the algebraic quadratic (24) for F implies that the branch locus of A is contained in the branch locus of F together with K_Ψ .*

Proof. Write $A = F + (A - F)$. The defining recurrences differ only in the ψ_k placement (Remark 10). Subtraction of (20) from (15) gives

$$a(k) - f(k) = \psi_k \left[Q a(k-1) + R \sum_{j+\ell=k-1} a(j) a(\ell) \right] - Q f(k-1) - R \sum_{j+\ell=k-1} f(j) f(\ell) - P\psi_k. \quad (105)$$

Summed into generating functions, the difference $A - F$ satisfies a linear inhomogeneous functional equation in $\mathcal{K}(\Psi_\mu)$ whose coefficients are meromorphic on $\tilde{X} \setminus K_\Psi$ and whose forcing terms are meromorphic on

$$\tilde{X} \setminus (K_F \cup K_\Psi) = \tilde{X} \setminus K_F \quad (\text{using } K_\Psi \subseteq K_F). \quad (106)$$

The analytic-implicit-function theorem applied at every point of $\tilde{X} \setminus K_F$ where the linearised operator

$$\partial_F[z R F^2 - (1 - z Q) F + P\Psi_\mu] = 2 z R F - (1 - z Q) \quad (107)$$

is nonzero produces a unique meromorphic solution $A - F$ on $\tilde{X} \setminus K_F$. The exceptional set where (107) vanishes is contained in K_F (by definition of branch locus). Therefore $A = F + (A - F)$ is meromorphic on $\tilde{X} \setminus K_F$. $\square$

9.4 Convergence in capacity (Stahl)

Theorem 27. *[Convergence in capacity] Let $\Phi \in \mathcal{A}$ with extremal compact K_Φ . Let $\{R_M\} = \{N_M/D_M\}$ be the diagonal Padé sequence built from the Taylor series of Φ at $z = 0$. Then $R_M \rightarrow \Phi$ in logarithmic capacity on every compact subset of $\tilde{X} \setminus K_\Phi$.*

Proof. The denominators D_M have all zeros in K_Φ up to a set of asymptotic capacity zero. This is the Nuttall–Stahl localisation: the zero-counting measure

$$\nu_M := M^{-1} \sum_{D_M(z)=0} \delta_z \quad (108)$$

converges weak-* to the equilibrium measure ν_{K_Φ} .

Step 1 (Hermite's formula). The Padé error satisfies

$$\Phi - R_M = \frac{\Pi_M(z)}{D_M(z)^2} \cdot \frac{1}{2\pi i} \oint_\gamma \frac{D_M(\zeta)^2 \Phi(\zeta)}{\Pi_M(\zeta)(\zeta - z)} d\zeta, \quad \Pi_M \text{ monic of degree } M, \quad (109)$$

where γ encircles K_Φ .

Step 2 (Capacity estimate). Outside K_Φ , the Green function $g(z, K_\Phi)$ controls

$$|D_M(z)|^{1/M} \longrightarrow e^{g(z, K_\Phi)} \text{cap}(K_\Phi) \quad \text{in capacity} \quad (110)$$

(Stahl's main lemma, proved by combining Bernstein–Walsh and the asymptotic distribution of zeros [StahlExtremal, StahlExtremalII]).

Step 3 (Vanishing in capacity). The error has M -th root behaviour

$$|\Phi(z) - R_M(z)|^{1/(2M)} \rightarrow e^{-g(z, K_\Phi)} \quad \text{in capacity,} \quad (111)$$

which is <1 off K_Φ , giving exponential vanishing in capacity.

The full proof of Steps 2–3 mirrors Stahl’s [StahlConvergence, StahlBranch] but, restricted to $\mathcal{A}$, simplifies because the discriminant Δ_Φ has only finitely many zeros and poles on the universal cover modulo K_Ψ , so the orthogonality measure of D_M has a smooth density off K_Φ and the standard equilibrium analysis applies directly. The polynomial Π_M in (109) plays the role of an auxiliary monic factor in Hermite’s contour formula and is unrelated to the orthogonal polynomials P_M of Theorem 17. $\square$

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